DevOps Series
Ansible Deployment *of* Monit

This is the 18th article in the DevOps series and it discusses the Ansible deployment of Monit, a free and open source utility for managing and monitoring processes, programs, files, directories and file systems on a *nix system.

Monit is a free and open source process supervision tool for *nix systems. It can also be used to monitor files and directories, and perform maintenance or repair tasks. The system status check can be done on the command line and viewed in a browser. It is written entirely in C and released under the AGPL 3.0 licence. In this 18th article in the DevOps series, we will learn to install and set up Monit for the system, as well as the SSH daemon and Nginx Web server monitoring.

Setting it up

A Debian 9 (x86_64) guest virtual machine (VM) using KVM/QEMU will be set up and monitored using Monit.

The host system is a Parabola GNU/Linux-libre x86_64 system and Ansible is installed using the distribution package manager. The version of Ansible used is 2.6.0, as indicated below:

```
$ ansible --version
ansible 2.6.0
  config file = /etc/ansible/ansible.cfg
  configured module search path = ['/home/guest/.ansible/
plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3.6/
site-packages/ansible
  executable location = /usr/bin/ansible
  python version = 3.6.5 (default, May 11 2018, 04:00:52)
[GCC 8.1.0]
```

The Ansible playbook and inventory file are created on the host system as follows:

```
ansible/inventory/kvm/
       /playbooks/configuration/
```

The *inventory/kvm/inventory* file contains the following code:

```
debian ansible_host=192.168.122.197 ansible_connection=ssh
ansible_user=debian ansible_password=password
```

The default Debian 9 installation does not have the *sudo* package installed. Log in to the VM and install the *sudo* package. The 'debian' user also requires *sudo* access:

```
root@debian:~# apt-get install sudo

root@debian:~# adduser debian sudo
Adding user `debian' to group `sudo'...
Adding user debian to group sudo
Done.
```

You should add an entry in */etc/hosts* file for the Debian VM as shown below:

```
192.168.122.197 debian
```

You can now test connectivity from Ansible to the Debian 9 VM using the following command:

```
$ ansible -i inventory/kvm/inventory debian -m ping
debian | SUCCESS => {
    "changed": false,
    "ping": "pong"
}
```

Installation

The Debian software package repository is first updated and then Monit is installed. The *net-tools* package is installed to provide the *netstat* command in the system. The Monit service is then started using *systemd*. The Ansible playbook for the above tasks is provided below, for reference:

```
---
- name: Install Monit
  hosts: debian
  become: yes
  become_method: sudo
  gather_facts: yes
  tags: [install]
```